

DIPOLE TYPE BY LENGTH

Compute first $\lambda=c/f$, then ratio l/λ		
l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D=1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\ \Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}}=199\ \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$
$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi} \frac{e^{-jkR}}{R} \sin \theta, \quad \vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density $l/\lambda < 1/50$; far field
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$
$S(R, \theta)$ avg. pwr dens. at (R, θ) (W/m ²). l antenna length (m, the rod itself). I_0 peak current in dipole (A). R distance from antenna to obs. point (m). θ angle from dipole axis (rad/°). $k=2\pi/\lambda$ wavenumber (rad/m). $\eta_0=120\pi \approx 377\ \Omega$ free-space impedance.
Max direction broadside $\theta_{\text{max}}=\pi/2$
Total radiated power integrate over 4π
$P_{\text{rad}} = \frac{\eta_0 \pi}{3} \left(\frac{I_0 l}{\lambda}\right)^2 \text{ (W)}$
Directivity exact $D_{\text{Hertz}}=1.5$ (1.76 dB)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R any l
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$
Normalized pattern dimensionless $F(\theta)=S(\theta)/S_{\text{max}}$
Quarter-wave $l=\lambda/4$ $\pi l/\lambda=\pi/4$, $\cos(\pi/4)=\sqrt{2}/2$
$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 = \frac{15 I_0^2}{\pi R^2}$
$S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$
Half-wave $l=\lambda/2$ $\pi l/\lambda=\pi/2$, shortcut form
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}$
$D=1.64$, $R_{\text{rad}}=73\ \Omega$, $\theta_{\text{max}}=\pi/2$.

COMMON ANGLES (deg ↔ rad, trig values)

°	rad	$\sin \theta$	$\cos \theta$	$\sin^2 \theta$	$\cos^2 \theta$	θ
0	0	0	1	0	1	
30	$\pi/6$	$1/2$	$\sqrt{3}/2$	$1/4$	$3/4$	
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	$1/2$	$1/2$	
60	$\pi/3$	$\sqrt{3}/2$	$1/2$	$3/4$	$1/4$	
90	$\pi/2$	1	0	1	0	
180	π	0	−1	0	1	

Convert: $\theta_{\text{rad}}=\theta_{\text{deg}}\cdot\pi/180$. Antenna formulas usually take θ in rad.

BEAM PARAMETERS — β, Ω_p, D

Step 1 — Normalize always start here
$F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)
Step 2 — HPBW β solve $F(\theta_{1/2}) = 0.5$, then $\beta = 2\theta_{1/2}$ for symmetric beams
Gaussian shortcut: $F=e^{-a\theta^2}=0.5 \Rightarrow \theta_{1/2}=\sqrt{\ln 2/a}$.
Step 3 — Pattern solid angle Ω_p units: sr
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$
No ϕ dep.: $\Omega_p=2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$.
Most integrals have NO closed form. Use Wolfram / numerical.
Wolfram: <code>2*pi*Integrate[F[t]*Sin[t],{t,0,pi}]</code>
Step 4 — Directivity D
$D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}}=10 \log_{10} D$
Alt. with two orthogonal HPBW's: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$.
Step 5 — Gain (if asked) $G=\xi D$ $\xi=\text{rad. eff.}$, $0 < \xi \leq 1$
$G = \xi D$ $G_{\text{dB}}=10 \log_{10} G$
Lossless / ideal antenna $\Rightarrow G=D$.

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	R_{rad} (Ω)
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	1.64	2.15	73
$l=\lambda$	2.41	3.82	199

Half-wave is the default “dipole antenna” unless stated otherwise.

EFFECTIVE AREA A_e

Definition antenna's RX cross section (m²)
$A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)
Physical cross section wire dipole, diameter d
$A_p = l d = \frac{\lambda}{2} d$ (half-wave)
Inverse: D from A_e
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)
Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.

R_{rad} , EFFICIENCY, POWER SPLIT

Radiation resistance Ω , equiv. ckt. load absorbing P_{rad}
$R_{\text{rad}} = \frac{2P_{\text{rad}}}{I_0^2} \quad R_{\text{loss}} = \frac{2P_{\text{loss}}}{I_0^2}$
Input power split P_t from generator
$P_t = P_{\text{rad}} + P_{\text{loss}} \quad R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}$
Radiation efficiency two equivalent forms
$\xi = \frac{P_{\text{rad}}}{P_t} = \frac{R_{\text{rad}}}{R_{\text{rad}}+R_{\text{loss}}}$ (unitless)

FRIIS TRANSMISSION FORMULA

Far-field link budget P_t tx, P_r rx, range R
$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t$ (W)
G_t, G_r are linear gains; $\lambda=c/f$; R in m; P_t, P_r in W.
Solve for R max range
$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$ (m)
Equivalent form using A_e recv. area form
$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$
Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.
Recipe 1. $\lambda=c/f$. 2. Convert dB gains to linear: $G=10^{G_{\text{dB}}/10}$.
3. If TX is “half-wave dipole” use $G_t=1.64$. 4. Plug into Friis.
Use linear G , not dB. Convert FIRST.

dB ↔ LINEAR

$G_{\text{dB}} = 10 \log_{10} G_{\text{lin}}$	$G_{\text{lin}} = 10^{G_{\text{dB}}/10}$
dB linear	dB linear
0 1	10 10
3 2	13 20
6 4	20 100
7 5	30 1000
−3 0.5	−10 0.1

Memorize: 3 dB ↔ ×2, 10 dB ↔ ×10. Add dB = multiply linear.

For E-field / amplitude use $20 \log$ not $10 \log$ (power ∝ E^2).

POWER FLOW IDENTITIES

$P_{\text{rad}} = S_{\text{max}} \Omega_p R^2 = \frac{4\pi R^2 S_{\text{max}}}{D} \text{ (W)}$
$S_{\text{max}} = \frac{P_{\text{rad}} D}{4\pi R^2} \text{ (W/m}^2\text{)}$
Isotropic radiator: $S=P_{\text{rad}}/(4\pi R^2)$ (no D factor).

SYMBOL REFERENCE

l — antenna length (m); $\lambda=c/f$ (m); $c=3\times 10^8$ m/s
I_0 — peak current (A); $\eta_0=120\pi \approx 377\ \Omega$
$k=2\pi/\lambda$ wavenumber (rad/m); R obs. dist (m)
$S(R, \theta)$ — power density (W/m ²)
$F(\theta)=S/S_{\text{max}}$ — normalized pattern (unitless)
β — half-power beamwidth (rad or deg)
Ω_p — pattern solid angle (sr)
$D=4\pi/\Omega_p$ — directivity (unitless); ξ rad. eff.; $G=\xi D$
$A_e=\lambda^2 D/(4\pi)$ — effective area (m ²); $A_p=ld$ phys. (m ²)
R_{rad} — radiation resistance (Ω); R_{loss} loss res. (Ω)
P_t, P_r — TX / RX power (W); G_t, G_r linear gains